



American International University-Bangladesh (AIUB)
Department of Computer Science
Faculty of Science & Technology (FST)

TAX COUNTING AND RETURN GENERATOR

PROJECT PROPOSAL

1.1 Background to the Problem

Taxes can be confusing and time-consuming for people and businesses. The problem is that manual tax calculations often lead to mistakes because tax rules are complicated and change a lot. This can cause problems like paying too much or too little in taxes, which can result in penalties and damage your reputation.

The main issue is that taxes have many rules, exceptions, and deductions, making it easy to make errors when calculating them by hand. Mistakes in tax calculations are a big deal because they can lead to legal trouble and financial problems for individuals and businesses. Taxes are also crucial for funding government services, so it's important to get them right.

To solve this problem, we need a smart tax calculator software. This software will make tax calculations easier, reduce mistakes, keep up with the latest tax rules, and help people and businesses manage their taxes correctly. This project is not just about personal convenience; it has a big impact on the economy and ensures that everyone follows the rules.

1.2 Solution to the Problem

For a tax calculation software, the target group of users typically includes individuals and businesses who need to accurately calculate and manage their taxes. Here's a breakdown of potential user groups and how they would benefit from the proposed tax calculation solution:

Individuals can use the app to easily calculate their income tax liabilities, deductions, and credits. The app can provide a user-friendly interface for entering personal financial information, ensuring that users are aware of potential tax savings.

Small business owners can utilize the app to calculate and track their business income, expenses, and tax obligations. The app may also assist in identifying eligible business deductions, simplifying the tax compliance process for entrepreneurs.

Tax professionals can benefit from using the app as a tool for efficiently managing client data, streamlining tax calculations, and staying updated on tax regulations. The app can enhance the accuracy and speed of tax preparation, allowing professionals to focus on providing valuable advice and strategic planning.

Overall, the tax calculation app aims to benefit users by providing a reliable, efficient, and user-friendly platform for managing various aspects of taxation. It can simplify the tax-related processes, enhance accuracy, and empower users with the information they need to make informed financial decisions.

The development of a tax calculation app can contribute to scientific results and knowledge in several ways, particularly in the field of financial technology (FinTech) and applied computer science. Here are some key contributions:

- Automation and Efficiency in Tax Compliance:
- Data Security and Privacy:
- User Experience and Human-Computer Interaction (HCI):
- Real-time Updates and Tax Regulation Changes:
- Machine Learning for Tax Prediction:
- Integration with Financial APIs and Services:

In summary, the tax calculation app project contributes to scientific results by advancing knowledge in automation, data security, user experience, real-time updates, machine learning, API integration, financial education, and technology scalability within the domain of financial technology and applied computer science.

Literature review-

As of my last knowledge update in January 2022, I don't have specific information on recent or ongoing studies on tax calculation apps. Therefore, I can provide a general outline of what a literature review might look like, incorporating common themes and issues in the field of tax technology. Please ensure to check the latest literature for any updates or new studies in this area.

- Automation in Tax Compliance:
- User Experience in Financial Applications:
- Security and Privacy in Financial Technology:
- Real-time Updates and Regulatory Compliance:
- Machine Learning in Financial Forecasting:
- Integration with APIs and Financial Services:
- Financial Education and Literacy
- Scalability and Cross-platform Compatibility:

Contributions and Extensions:

- Innovative Features:
- Comprehensive User-Centric Approach:
- Advanced Security Measures:

- Unique Integration Strategies:
- Effective Real-time Updates:
- Incorporation of Machine Learning:
- Emphasis on Financial Literacy:
- Scalable Architecture:

In our study, we underscore the distinctive features inherent in our tax calculation app, highlighting its innovative approach to addressing critical gaps within the existing literature on financial technology and tax applications. By meticulously emphasizing the uniqueness of our solution, we aim to make valuable contributions to the field.

2. SOFTWARE DEVELOPMENT LIFE CYCLE

2.1 Process Model

The process model which will be suitable for our project is **SCRUM**.

It will be appropriate for our project because it is a lightweight management practice. Scrum is a popular agile framework which offers numerous benefit that emphasizes flexibility, cooperation, and iterative development.

Scrum will allow us to make changes throughout the development process because this is crucial for a tax counter software, as tax regulations and requirements may change over time. Scrum enables us to adapt these changes quickly. Unlike, the other process such as water fall model where it is not possible to make changes whenever changes are needed because this are sequential process model once it completed it cannot be revisited or changed. As tax rate are change frequently, we need to update our system regularly that's why scrum process model is best for our software.

2.2 Project Role Identification and Responsibilities

- 1) **Product Owner:** the person who will be responsible for the product. The product owner defines and prioritize the product backlog, ensuring that the development team is working on the most valuable and essential features.
- 2) **Stakeholders:** Stakeholders are individuals or groups who are affected by the outcome of the project. They provide feedback, set expectations, and ensure that the product being developed aligns with business goals and user needs. Their input is crucial during review sessions and planning.

- 3) **Scrum Master:** They will be the middle man. He will assist the product owner and the development team. The Scrum master will help the Product owner by suggesting techniques to manage the product log and the development team by coaching the team to self-organize and remove roadblocks.
- 4) **Development Team:** A development team is composed of professionals who do the hands- on work of completing the tasks in a Scrum sprint. They Help in sprint planning and goal setting, lend expertise to program, design, or improve products Use data to find best practices for development and also Test products and prototypes, plus other forms of quality assurance.
- 5) **Quality Assurance (QA) Engineer:** The QA Engineer ensures that the product meets the required standards and is free from defects. They write and execute test cases, perform regression testing, and work closely with the Development Team to identify and resolve issues early in the sprint cycle.

3. SOFTWARE REQUIREMENT ANALYSIS

3.1 Functional Requirements:

1. Login

- a) Enable users to securely access the system using their personal identification (username or email) and a password.
- b) If someone has trouble logging in, send a security code to their email for verification. Priority level: High.

Precondition: User must be registered with valid access information.

2. Registration

- a) Users access the registration page through the registration button.
- b) Users enter necessary details like username, phone number, email, and password into the specified sections on the registration form.
- c) The system verifies the provided data to confirm it adheres to specified criteria, including proper email format, password complexity, and completion of all mandatory fields.

d) **Optional Email Verification:** If activated, a confirmation email with a distinct activation link is dispatched to the email provided during registration. Users must click on this link to activate their account and access the platform.

Priority level: High.

Precondition: To activate the account, it is necessary to supply a distinctive email address and adhere to the specified password criteria.

3) Role Based Access

a) Identify distinct roles that users can have in the system. Standard roles may include administrators and customers.

b) Allow administrators to create, modify and delete.

c) Allow customers to update their personal information and customize their profiles. Priority level: High.

Precondition: The role must be matched with the information stored in the database.

3) Tax Calculator

a) Allow users to calculate the amount of tax imposed on their property.

b) Users enter different type of their property type and amount of the property (land, car, income, business, gold).

c) The system calculates the provides property and amount, then it will generate the amount of tax imposed in each type of property.

Priority level: High.

Precondition: Users must have enough property to impose tax

4) E-tin registration

- a) Allow users to registration E-tin, who does not have E-tin form registration.
- b) Users enter necessary details like username, phone number, email, and password into the specified sections on the registration E-tin form.
- c) The system verifies the provides data to confirm the registration.

Priority level: High.

Precondition: User must be registered with valid information.

5) Tax Return Generator

- a) Allow users to creates a tax return file.
- b) Users input e-tin number, NID number address and other information.
- c) The system verifies the provides information along with the tax calculated form the tax calculator and creates a tax return file.
- d) Users can download the pdf copy of tax return file.

Priority level: High.

Precondition: Tax must be imposed to the users.

3) Tax Return Submission

- a) Allow users to submit their tax return file.
- b) Users will get a link to submit the pdf generated by tax return generator. Priority level: High.

Precondition: Understanding how to submit files or upload files.

4) **Tax Zone**

- a) Allow users to know the nearest tax return submission office in their area.
- b) Users can choose their area and access area-specific tax office's address and contact number.

Priority level: Medium.

Precondition: The system should have accurate geographical data and database of tax office with corresponding addresses and contact information.

5) **Notification**

- a) Administrators can send notification to users informing them of scheduled system updates, maintenance downtime or any other system related announcements.
- b) Users receive a notification when tax rates are updated to the system or any kind of system update.
- c) Users can customize their notification preferences based on their specific needs and preferences, choosing the types of notifications they want to receive and how they want to receive them (email, in-app notification, mobile push notifications).

Priority level: Medium.

Precondition: Existence of a user authentication system and channels for delivering notifications.

6) **Forget Password**

- a) Allow users to reset their password by providing a "Forgot Password" link on the login page.
- b) Upon clicking the "Forgot Password" link, prompt users to enter their email address associated with their account.
- c) Send a password reset link to the user's email address with instructions on how to reset their password.
- d) Ensure that the password reset link expires after a specified time period for security purposes.
- e) Implement security measures such as CAPTCHA or multi-factor authentication to

prevent unauthorized password resets.

Priority: High.

Precondition: Users must have a registered email address associated with their account for password reset.

7) **Two-Factor Authentication (2FA)**

- a) Provide an option for users to enable two-factor authentication (2FA) for an extra layer of security during login.
- b) Users must authenticate with a second factor (e.g., SMS code, email code, or authenticator app).
- c) Allow users to enable or disable 2FA from their profile settings.

Priority level: High

Precondition: User account must be properly registered with an email or phone number.

8) **User Profile Management**

- a) Allow users to view and update their personal information (name, email, phone number, etc.) in their profile.
- b) Users can upload and update their profile picture.
- c) Enable users to change their password (with the necessary verification and security measures).

Priority level: High

Precondition: User must be logged in to access their profile

9) **Transaction History**

- a) Allow users to view their historical transaction records related to tax payments, submissions, and other relevant activities.
- b) Provide filtering and search functionality for users to easily navigate through their transaction history.
- c) Allow users to download the transaction history in formats such as CSV or PDF.

Priority level: Medium

Precondition: User must have performed at least one transact

10) Tax Payment Gateway Integration

- a) Allow users to pay their calculated tax online through an integrated payment gateway (e.g., credit card, bank transfer, mobile payments).
- b) Display the total amount to be paid, along with any applicable payment fees or charges.
- c) Provide users with a payment receipt upon successful transaction.

Priority level: High

Precondition: Payment gateway integration must be set up, and user must have tax due for payment.

11) Audit and Logging

- a) Keep detailed logs of all actions performed by users and administrators (login attempts, registration, updates to profiles, tax submissions).
- b) Provide administrators with access to audit logs for security and tracking purposes.
- c) Implement alerting mechanisms for suspicious activities, such as multiple failed login attempts.

Priority level: High

Precondition: System must support log tracking and alert mechanisms.

12) Help and Support Center

- a) Provide users with access to a help center or FAQ section where they can find answers to common questions.
- b) Enable users to contact support via email, phone, or live chat for further assistance. Priority

level: Medium

Precondition: A helpdesk or support system must be in place.

13) Data Backup and Recovery

- a) Regularly backup all user data, including tax-related information, transaction history, and system settings.
- b) Implement a recovery process to restore the system in case of failure or data loss. Priority level: High

Precondition: Backup systems and recovery procedures must be established.

3.2 Non-functional requirements:

1. Usability: The system responds to user inputs within 2 seconds.

2. Reliability: The system should be available 99.9% of the time.

3. Security: Only authorized users have access to sensitive data or functionalities.

Communication between the client and server is encrypted by using SSL.

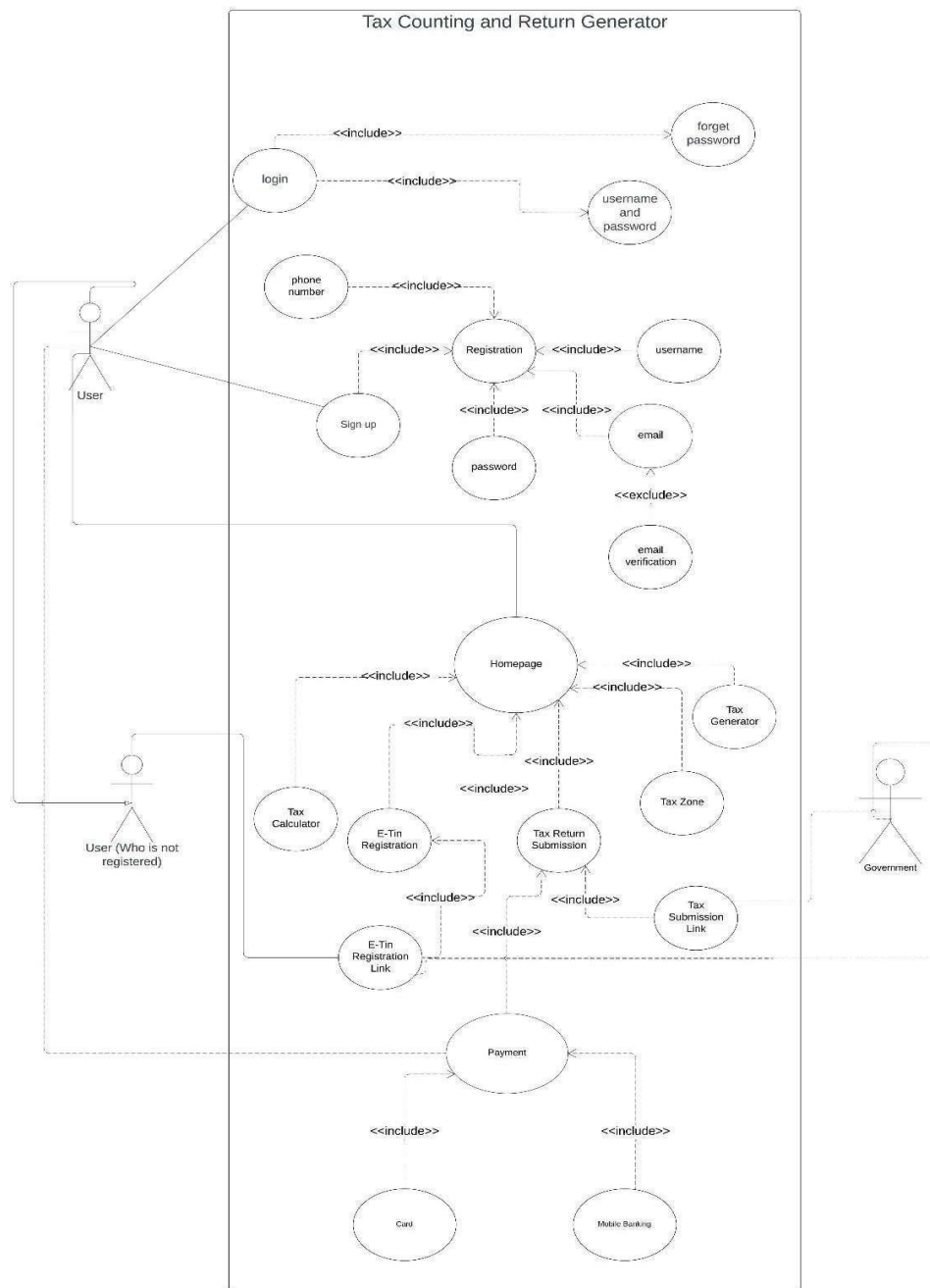
4. Maintainability: Code is well-documented and follow coding standards. The system architecture allows for easy modifications and updates.

5. Compatibility: The application is compatible with the latest versions of major web browsers. The software can run on Windows, macOS, and Linux operating systems.

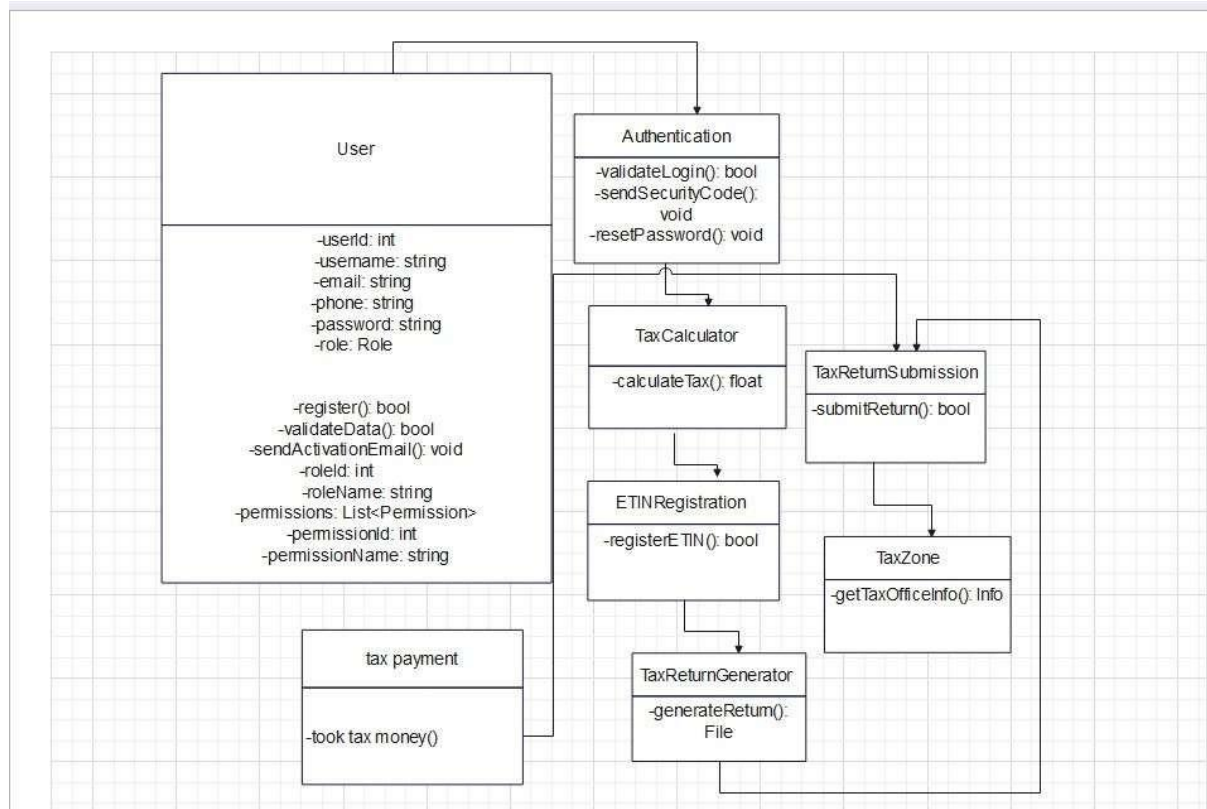
6. Interoperability: The system can integrate seamlessly with external payment gateways. The software support industry-standard data exchange formats

1. SOFTWARE DIAGRAMS

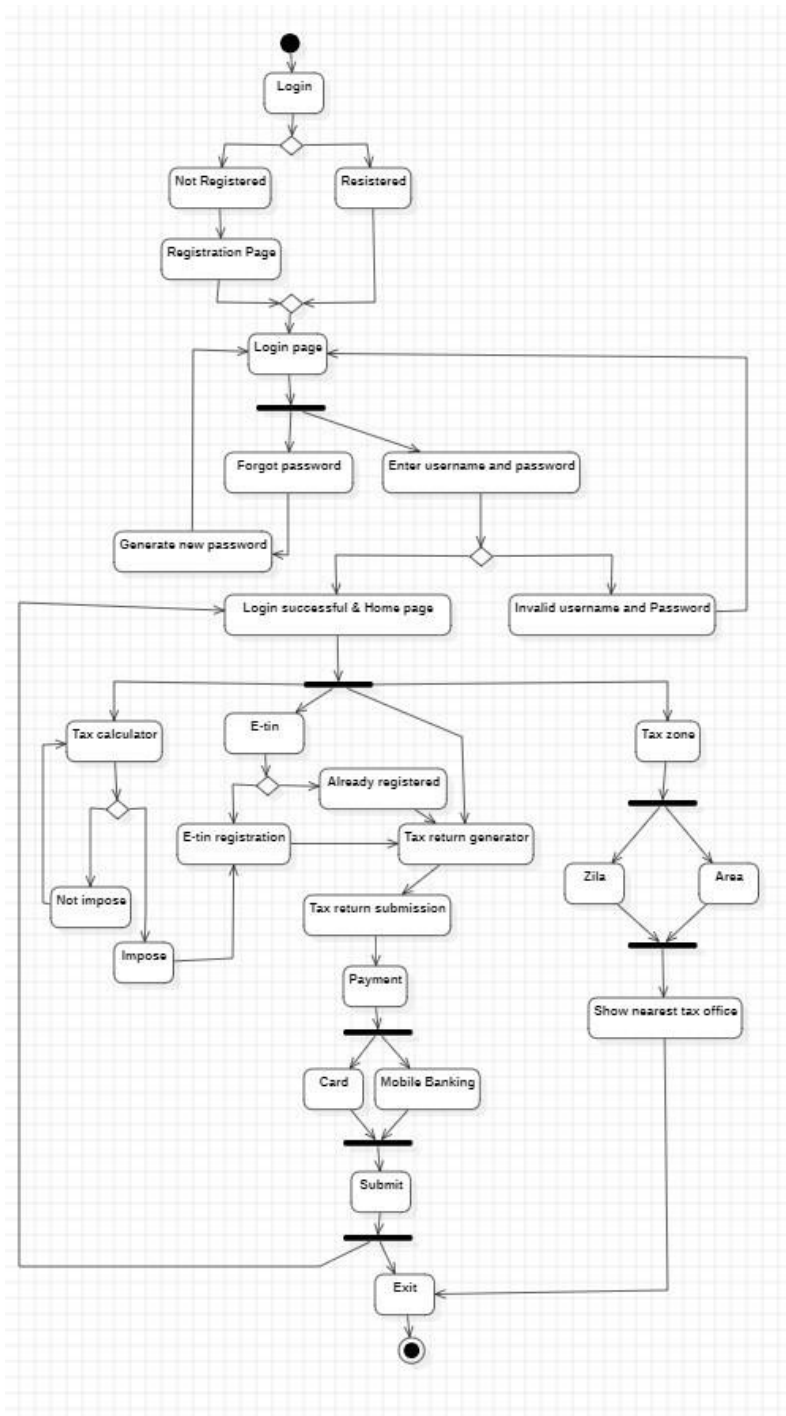
USE CASE DIAGRAM



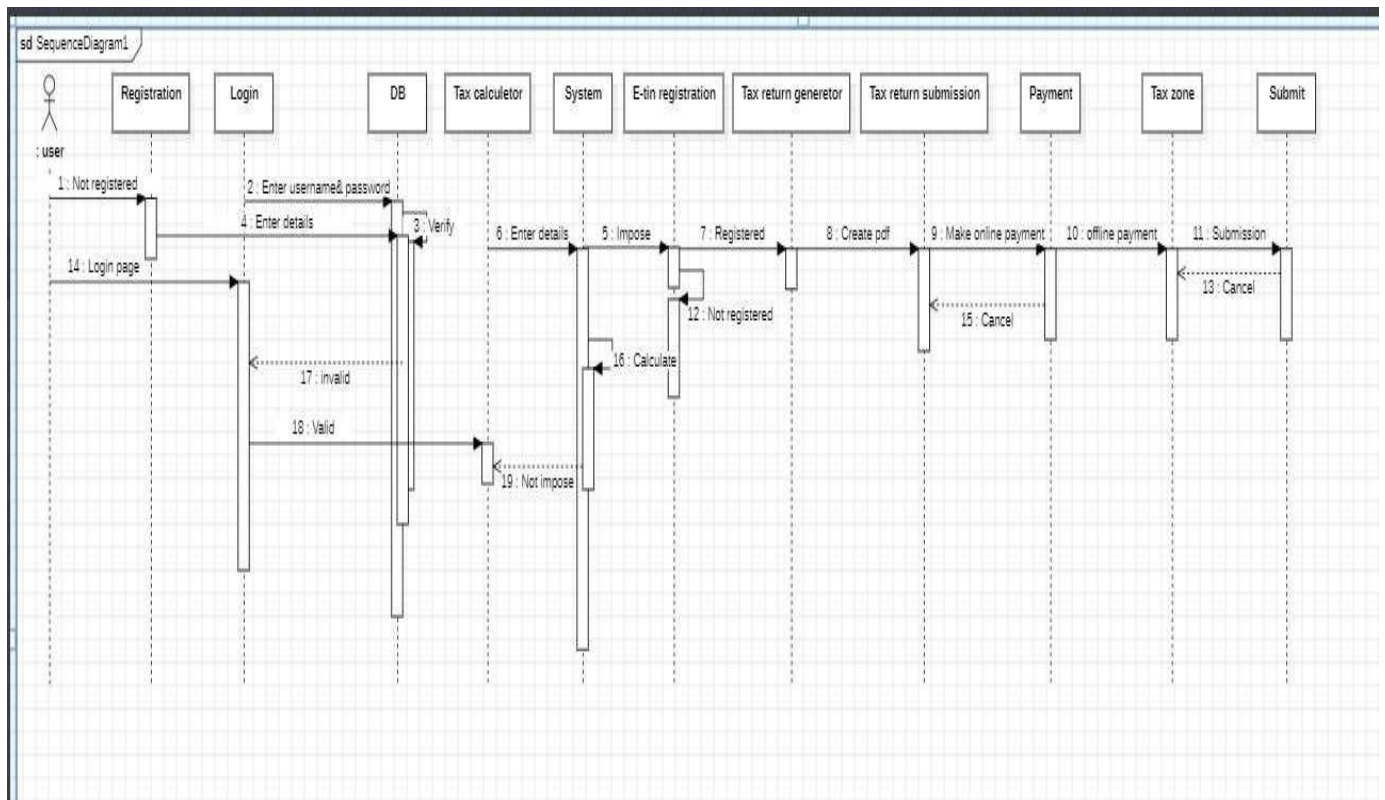
CLASS DIAGRAM



ACTIVITY DIAGRAM



SEQUENCE DIAGRAM



5. Software Design:

Log in



TAX COUNTING & RETURN GENERATOR

Login





[Log In →](#)

[Forget Password?](#)

OR

[Sign Up →](#)

Registration



Registration













[Register →](#)

Homepage



Homepage

Tax Calculator

e-TIN Registration

TAX Submission

TAX Zone

TAX Generator

Tax Calculator



TAX Calculator

Source of Income	Taka
1. Income from employment	_____ ট
2. Income from rent	_____ ট
3. Income from agriculture	_____ ট
4. Income from business	_____ ট
5. Capital income	_____ ট
6. Income from financial assets	_____ ট
7. Income from other sources	_____ ট
8. Taxable income arising abroad	_____ ট

Total Income: _____ ট

Total TAX: _____ ট

Submit

Tax Rate

*Tax on gross income up to first Tk. 3,50,000

0%

*Tax on gross income up to next Tk. 1,00,000

5%

*Tax on gross income up to next Tk. 3,00,000

10%

*Tax on gross income up to first Tk. 4,00,000

15%

*Tax on residual gross income

20%

Tax Return Generator



Tax Return Generator



Enter e-TIN number



Enter NID number



Enter email address

Generate Tax Return →

View TAX Return

Download PDF

Tax Return Submission



Tax Return Submission

Total Tax amount: _____ ₳



Payment

How would you like to pay?

☐

bKash



☐

Nagad



☐

Upay



☐

Credit or Debit Card



☐

Other online payment methods

>

Confirm

Tax Zone

Forgot Password



Tax Zone

TAX Zone-9
Address: 32 Gareeb-e-Nawaz Ave,
Dhaka-1230
Telephone: 02-36583957
Phone: 01724931119
Website: [http://
www.taxzone9dhaka.gov.bd/](http://www.taxzone9dhaka.gov.bd/)



Forgot password

e –Tin Login



e-Tin Login

Welcome to
Taxpayer's Identification
Number (TIN)
Registration / Cancellation



Enter User ID



User Password

[Forget Password?](#)

Login →

Do not have User ID?

Register

e-TIN Verification

e –Tin Registration



e-Tin Registration

User ID:

Password:

Retype Password:

Country ▼

Mobile: +880

Email Address:

Register

e-Tin Profile



Profile



Name: Sakib Khan

Mobile: 01643834638

e-TIN No: 6734873755

Actions

 Edit Profile Info

 Change Password

 Change Language

Logout

6. Test Case:

Project Name: tax counting and return generator		Test Designed by: Parvez Ahmed Joy		
Test Case ID: Test Priority (Low,Medium,High): High		Test Designed date:5/11/2025 Test Executed by:		
Module Name: Log in		Test Execution date:		
Test Title: Log in				
Description: This test case used for log in .				
Precondition (If any):				
Test Steps 1. Firstly you have to download this app 2. Then put their username. Or email (if they have) 3: after giving username then they have to put their password. 4: If find any issues directly sent a security code in the phone number or email. 5: Then if verify is successfully then the user will be able to access their	Test Data 1.Number:01***** Or email(aro*****@gmail.com) 2. Username: Text1234 3. Password: Valid2345	Expected Results	Actual Results	Status (Pass/Fail)
		Information's are saved in the database		

accounts				
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Post Condition: The user account is created and can be used to log in. The database updates with the user's information and check their validation.

Project Name: tax counting and return generator			Test Designed by: Parvez Ahmed Joy	
Test Case ID: Test Priority (Low,Medium,High): High			Test Designed date:5/11/2025 Test Executed by:	
Module Name: Registration			Test Execution date:	
Test Title: Registration				
Description: This test case used for registration				
Precondition (If any):				
Test Steps 1. Users access registration button. 2. Users enter necessary details like username, phone number, email, and password. 3. The system verified the person 4. Optional Email Verification: If activated, a confirmation email.	Test Data 1.Number:01***** Or email(aro*****@gmail.com) 2.Username: Text1234 3.Password: Valid2345	Expected Results	Actual Results	Status (Pass/Fail)
		Information's are saved in the database		
Post Condition: The user account is created and can be used to registration. The database updates with the user's information and check their validation.				

Test Case

Project Name: tax counting and return generator			Test Designed by: Md. Shihab Hossain Khan	
Test Case ID: Test Priority (Low,Medium,High): High			Test Designed date:5/11/2025 Test Executed by:	
Module Name: Role Based Access			Test Execution date:	
Test Title: Role Based Access				
Description: This test case used for the main role of this software				
Precondition (If any):				
Test Steps 1. Identify distinct roles that users can have in the system. 2. Allow administrators to create, modify and delete. 3. Allow customers to update their personal information and customize their profiles.	Test Data 1.Number:01***** Or email(aro*****@gmail.com) 2. Username: Text1234 3. Password: Valid2345	Expected Results	Actual Results	Status (Pass/Fail)
		Information's are saved in the database		
Post Condition: The user account is created and can be used to Role Based Access. The database updates with the user's information.				

Test Case

Project Name: tax counting and return generator			Test Designed by: Md. Shihab Hossain Khan	
Test Case ID: Test Priority (Low,Medium,High): High			Test Designed date:5/11/2025 Test Executed by:	
Module Name: Tax Calculator			Test Execution date:	
Test Title: Tax Calculator				
Description: This test case used for the tax calculating. What amount of money you have to paid				
Precondition (If any):				
Test Steps 1. Allow users to calculate the amount of tax imposed on their property. 2. Users enter different type of their property type and amount of the property (land, car, income, business, gold). 3. The system calculates the provides property and amount, then it will generate the amount of tax imposed	Test Data 1.Number:01***** Or email(aro*****@gmail.com) 2. Username: Text1234 3. Password: Valid2345 4: amount of money:22*****	Expected Results	Actual Results	Status (Pass/Fail)
		Information's are saved in the database		
in each type of property.				

Post Condition: The user account is created and can be used to tax calculator. The database updates with the user's information .and counting money				

Test Case

Project Name: tax counting and return generator	Test Designed by: Fariha Meem
Test Case ID: Test Priority (Low,Medium,High): High	Test Designed date:5/11/2025 Test Executed by:
Module Name: E-Tin Registration Link	Test Execution date:
Test Title: E-Tin Registration Link	
Description: This test case used for the tax calculating. What amount of money you have to paid	
Precondition (If any):	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Allow users to registration E-tin, who does not have E-tin form registration. 2. Users enter necessary details like username, phone number, email, and password into the specified sections on the registration E-tin form. 3. The system verifies the provides data to confirm the registration.	1.Number:01***** Or email(aro*****@gmail.com) 2. Username: Text1234 3. Password: Valid2345 4: amount of money:22***** 5: E-TIN pdf:	Information's are saved in the database		
Post Condition: The user account is created and can be used to tax calculator. The database updates with the user's information .and counting money				

Test Case

Project Name: Tax Counting and Return Generator			Test Designed by: Fariha Meem	
Test Case ID: Test Priority (Low, Medium, High): High			Test Designed date:5/11/2025 Test Executed by:	
Module Name: Tax Return Generator			Test Execution date:	
Test Title: Tax Return Generator				
Description: This test case used for tax return generator				
Precondition (If any):				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
		Information's are saved in the database		
1. Allow users to creates a tax return file. 2. Users input e-tin number, NID number address and other information. 3. The system verifies the provides information along with the tax calculated form the tax calculator and creates a tax return file. 4. Users can download the pdf copy of tax return file.	1.Number:01***** Or email(aro*****@gmail.com) 2. Username: Text1234 3. Password: Valid2345			
Post Condition: The user account is created and can be used to tax return generator. The database updates with the user's information and provide pdf copy of tax return file.				

Test Case

Project Name: Tax Counting and Return Generator			Test Designed by: Sabbir Hossain	
Test Case ID: Test Priority (Low, Medium, High): High			Test Designed date:5/11/2025 Test Executed by:	
Module Name: Tax Return Submission			Test Execution date:	
Test Title: Tax Return Submission				
Description: This test case used for tax return submission				
Precondition (If any):				
Test Steps 1. Allow users to submit their tax return file. 2. Users will get a link to submit the pdf generated by tax return generator.	Test Data 1.Number:01***** Or email(aro*****@gmail.com) 2.Username: Text1234 3.Password: Valid2345	Expected Results	Actual Results	Status (Pass/Fail)
		Information's are saved in the database		
Post Condition: The user account is created and can be used to tax return generator. The database updates with the user's information and provide a link to submit the pdf.				

Test Case

Project Name: Tax Counting and Return Generator			Test Designed by: Sabbir Hossain	
Test Case ID: Test Priority (Low, Medium, High): High			Test Designed date:5/11/2025 Test Executed by:	
Module Name: Tax Zone			Test Execution date:	
Test Title: Tax Zone				
Description: This test case used for tax zone				
Precondition (If any):				
Test Steps 1. Allow users to know the nearest tax return submission office in their area. 2. Users can choose their area and access areaspecific tax office’s address and contact number.	Test Data 1.Number:01***** Or email(aro*****@gmail.com) 2.Username: Text1234 3.Password: Valid2345	Expected Results	Actual Results	Status (Pass/Fail)
		Information’s are saved in the database		
Post Condition: The user account is created and can be used to tax zone. The database updates with the user’s information and provide all contact information of nearest tax submission office.				

Test Case

Project Name: Tax Counting and Return Generator		Test Designed by: Nurul Alam	
Test Case ID: Test Priority (Low, Medium, High): High		Test Designed date:5/11/2025 Test Executed by:	

Module Name: Notification			Test Execution date:	
Test Title: Notification				
Description: This test case used for tax zone				
Precondition (If any):				
Test Steps 1. Administrators can send notification to users informing them of scheduled system updates, maintenance downtime or any other system related announcements. 2. Users receive a notification when tax rates are updated to the system or any kind of system update. 3. Users can customize their notification preferences based on their specific needs and preferences, choosing the types of notifications they want to receive and hoe they want to receive them (email, in-app notification,	Test Data 1.Number:01***** Or email(aro*****@gmail.com) 2. Username: Text1234 3. Password: Valid2345	Expected Results	Actual Results	Status (Pass/Fail)
		Information's are saved in the database		

mobile push notifications).				

Post Condition: The user account is created and can be used to notification. The database updates with the user’s information and provide notification specific needs and preferences.

Test Case

© MMH

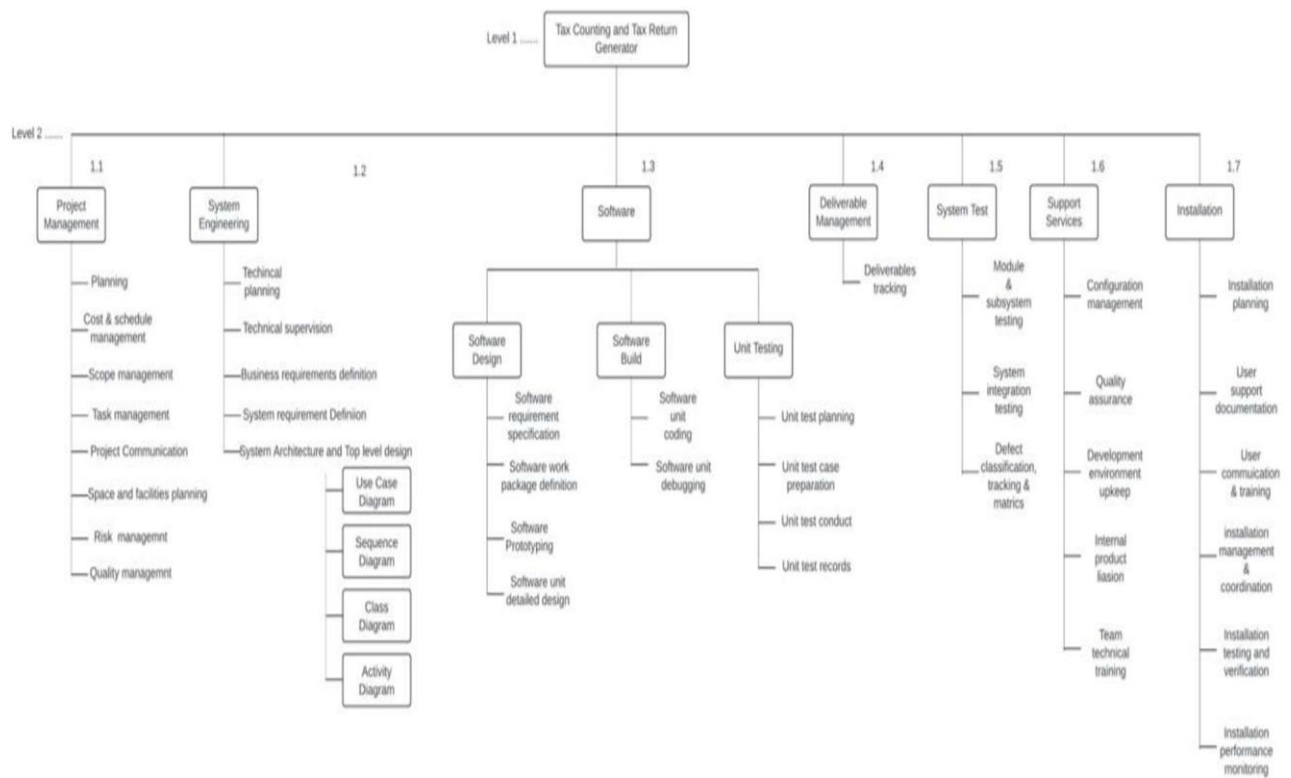
Project Name: tax counting and return generator	Test Designed by: Nurul Alam
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Test Case ID:		Test Designed		
Test Priority (Low, Medium, High): High		date:5/11/2025		
Module Name: Forget Password.		Test Executed by:		
Test Title: Forget Password.		Test Execution date:		
Description: This test case used for tax zone				
Precondition (If any):				
f) Allow users to reset their password by providing a "Forgot Password" link on the login page. g) Upon clicking the "Forgot Password" link, prompt users to enter their email address associated with their account. h) Send a password reset link to the user's email address with instructions on how to reset their password. i) Ensure that the password reset link expires after a specified time period for security purposes.	Test Data 1.Number:01***** Or email(aro*****@gmail.com) 2.Username: Text1234 3.Password: Valid2345	Expected Results	Actual Results	Status (Pass/Fail)
		Information's are saved in the database		

j) Implement security measures such as CAPTCHA or multi-factor authentication to prevent unauthorized password resets.				
Post Condition: The user account is created and can be used to forget password. The database updates with the user's information.				

Test Case

7. WBS:



8. Effort Estimation :

Effort Estimation:

Our project is small, simple software project in which a small team with good application experience work in software development. As a result, the project is considered organic.

Constructive Cost Model (COCOMO):

Our SLOC (Source Line of Code) is 18000

For Organic;

Coefficient=2.4, P=1.05, T=0.38

So, Effort=PM=Coefficient*(SLOC/1000) ^p

$$=2.4(18000/1000)^p$$

$$=2.4*20.79$$

$$=49.91 \text{ (WBS)}$$

Development time = $2.5*(PM)^T$

$$=2.5*4.418$$

$$=11.04 \text{ weeks}$$

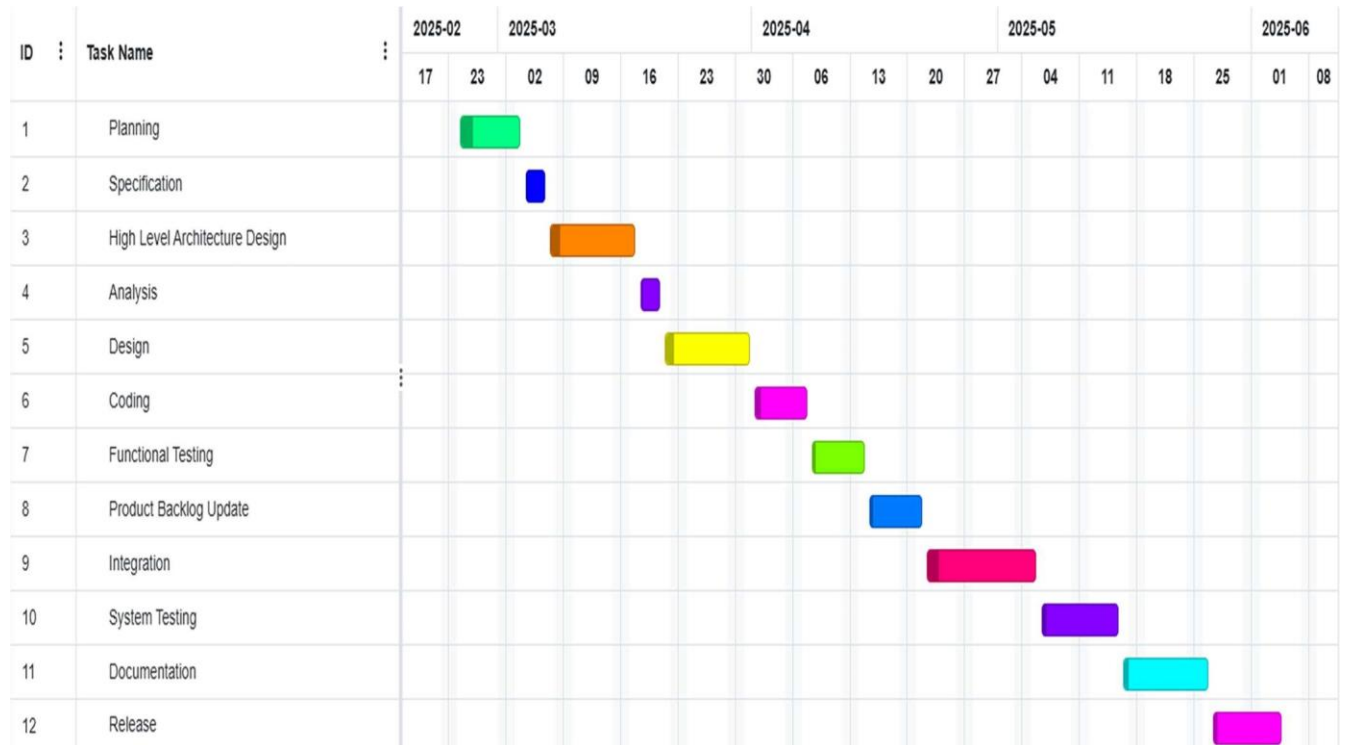
Required people = PM/DM

$$=49.91/11.04$$

$$=4.52 \sim 5 \text{ people}$$

So, the Required Development time for our 11.04 weeks or almost 3 months and required 5 people for the development of the project.

9. Timeline Chart :



<https://www.onlinegantt.com>

10. EVA

Task	Planned Effort	Actual Effort
1	14.0	15.0
2	9.0	6.5
3	9.0	10.0
4	7.0	7.0
5	7.0	14.5
6	10.0	8.0
7	8.0	8.5
8	8.0	7.0
9	9.0	12.0
10	6.0	3.0
11	9.0	13.5
12	8.0	9.0
13	8.0	-
14	8.0	-
15	8.0	-
16	6.0	-

BCWP = 104.0

BCWS = 104.0

ACWP = 114.0

Given total task = 38

Effort Estimated = 396 person-day

- $BAC = 396.0$
- $SPI = BCWP/BCWS$
 $= 104.0/134.0$
 $= 0.776119$
- $SV = BCWP - BCWS$
 $= 104.0 - 134.0$
 $= -30 \text{ person-day}$
- $CPI = BCWP/ACWP$
 $= 104.0/114.0 = 0.91$
- $CV = BCWP - ACWP$
 $= 104.0 - 114.0$
 $= -10 \text{ person-day}$

- % schedule for completion = $BCWS/BAC$
 $= 134.0/396.0$
 $= 33.84\%$ [% of work scheduled to be done at this time]
- % complete = $BCWP/BAC$
 $= 104.0/396.0$
 $= 26.26\%$ [% of work completed at this time]

11. Risk Management:

Risk Management:

Risk Description	Impact	Response	Owner	Notes
1. Data security breaches	High (Catastrophic)	Implement encryption, access controls, security audits, and compliance checks	QA Engineer	Conduct penetration testing and staff training.
2. Competitive pressure	Medium (Critical)	Monitor market trends, prioritize unique features, and accelerate development.	Product Owner	Perform SWOT analysis to stay competitive.
3. Legal liability (non-compliance)	High (Catastrophic)	Regular legal reviews, compliance checks, and tax algorithm validation.	Product Owner	Consult tax experts for accuracy
4. Large number of users than planned	Medium (Critical)	Design for scalability, conduct load testing, and use cloud-based solutions.	Development Team	Monitor server performance post-launch.

5. Less reuse than planned	Medium (Marginal)	Identify reusable modules early, document thoroughly, and allocate buffer time.	Development Team	Track reuse metrics in sprints
6. Staff inexperienced	High (Critical)	Provide training, pair programming, mentorship, and hire consultants if needed.	Scrum Master	Conduct skill gap analysis upfront.
7. Support and maintenance challenges	Medium (Marginal)	Allocate maintenance budget, document code rigorously, and automate testing.	QA Engineer	Plan long-term support in project scope.
8. Technology won't meet expectations	High (Critical)	Prototype early, evaluate alternatives, and involve tech leads in tool selection.	Development Team	Use "spikes" (research tasks) in Scrum.
9. High staff turnover	High (Critical)	Cross-train team, document processes, offer retention incentives.	Scrum Master	Monitor team morale and workload.

How to reduce risks:

Data Security	To protect sensitive tax data, strong encryption and role-based access controls should be used. Regular audits and secure coding practices must be enforced, along with staff training on data protection.
Competitive Pressure	Stay ahead by monitoring competitors, focusing on unique features like automatic tax updates, and delivering a smooth user experience. Agile methods help quickly adapt to market changes.
Legal Liability	Ensure compliance with tax laws through expert consultation, dynamic tax updates, and full documentation. This reduces the risk of legal issues due to incorrect calculations or outdated rules.
Large Number of Users	Use scalable cloud infrastructure and conduct regular load testing. Optimize performance to handle unexpected user growth without compromising system speed or stability.
Less Reuse Than Planned	Encourage modular development and thorough documentation to support code reuse. Regular code reviews help identify reusable components early in the process.
Staff Inexperience	Provide training and assign mentors to junior developers. Encourage collaboration and continuous learning to improve productivity and reduce errors.

Support and Maintenance	Establish a dedicated support team, use monitoring tools, and define clear response times. Maintain the software through regular updates and issue tracking.
Technology Expectations	Prototype early and involve stakeholders in choosing technologies. Plan for alternatives in case tools or frameworks fail to meet performance or feature needs.
Staff Turnover will be high	Promote a positive work environment, offer growth opportunities, and cross-train team members. Keep documentation up to date to reduce dependency on individuals.

Rubric for Project Assessment (CO3)

Criteria	Marks distribution (Max 3X5= 15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Selection of Software Engineering Models	Does not articulate a position or argument of choosing appropriate model. Does not present any evidence to support the arguments for the choice of the model	Articulates a position or argument for choosing models that is unfocused or ambiguous. Presents incomplete/vague evidence to support argument for model choice	Articulates a position or argument of choosing models that is limited in scope. Does not present enough evidence to support the argument for the choice of the model	Clearly articulates a position or argument for the choosing software engineering models. Presents sufficient amount of evidence to support argument for the model selection	
Role identification and Responsibility Allocation	The project has poor project management plans for identifying roles and assigning the responsibilities	Identify few roles in the project management where some of the roles are left alone with any project responsibilities	Identify most of the roles in the project management and assign their responsibilities	Well planned project with proper role identification and responsibility allocation in the project management activities	
Impact identification					
Formatting and Submission	Project report is not complete and Several errors in spelling and grammar. Present a Confusing organization of concepts, supporting arguments, and real-life example. Sentences rambling, and details are repeated.	Some errors in spelling and grammar. Some problems of organizing the answer in a logical order of defining, elaborating, and providing real-life examples.	Few errors in spelling and grammar. Presents most of the details in a logical flow of organization in definition, details, and example.	Project report is complete and No errors in spelling and grammar. Consistently presents a logical and effective organization of definition, details, and real-life example of the topic.	
Acquired marks:					
CO Pass / Fail:					

Rubric for Project Assessment (CO4)

Marking Criteria	Marks Distribution (Maximum 3X5=15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Project Planning	No background information regarding the project is given; project goals and benefits are missing.	Insufficient background information is given; project goals and benefits are poorly stated	Sufficient background information is given; the purpose and goals of the project are explained.	Thorough and relevant background information is given; project goals are clear and easy to identify.	
Effort Estimation and Scheduling	Student vaguely discuss the impact of societal, health, safety, legal and cultural issues in their project	Student provided with partial relevance to the impact of societal, health, safety, legal and cultural issues in their project	Student fairly provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	Student comprehensively provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	
Risk Management	Ambiguous representative example.	Partially identify / indicate towards real-life example.	Real-life example is fairly connected towards the definition.	Comprehensively defend with real life example.	
Acquired Marks:					
CO Pass / Fail:					

CO5 [PO-i-2]: Perform as an effective team member or leader in diverse team settings and solve multi-disciplinary problems in computer science and engineering domain.

Assessment Attribute/Criteria	Missing/ Incorrect (0)	Inadequate (1)	Satisfactory (2)	Excellent (3)
Taking responsibility	Does not perform assigned tasks; often misses meetings and, when present, does not have anything	Partially performs all assigned tasks; attends meetings irregularly and occasionally	Performs all assigned tasks; attends meetings regularly and usually participates effectively.	Performs all tasks very effectively; attends all meetings and participates enthusiastically; very reliable.

	constructive to say; relies on others to do the work;	participates and hence not reliable;	generally reliable;	
Contributions	Never provides useful ideas when participating in a group discussion	Rarely provides useful ideas when participating in a group discussion	Sometimes provides useful ideas when participating in a group discussion	Routinely provides useful ideas when participating in a group discussion
Collaboration and Ability to Compromise	Not cooperative, unable to compromise and disrupts the team process.	Sometimes cooperative, and rarely displays a positive attitude.	Usually cooperative, able to compromise and generally display positive attitude.	Always cooperative. Willingness to compromise. Always display positive attitude.
Valuing other team members (Working with others)	Often argues with teammates; doesn't let anyone else talk; occasional personal attacks and "put-downs"; wants to have things done his way and does not listen to alternate approaches.	Seldom listens to others' points of view; occasionally behaves in an oppressive manner; tries to force their own ideologies on other.	Generally, listens to others' points of view; always uses appropriate and respectful language; tries to make a definite effort to understand others' ideas.	Always listens to others and their ideas; helps them develop their ideas while giving them full credit; always helps the team reach a fair decision.